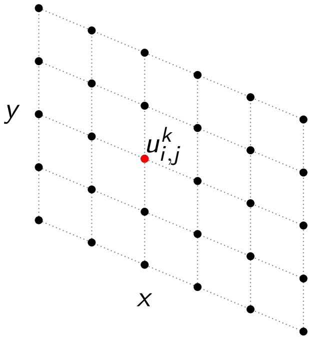
The Alternating Direction Implicit (ADI) Method

A Numerical Approach for Solving 2D Diffusion Equations

$$\frac{\partial u}{\partial t} = D\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

Discretize the Domain



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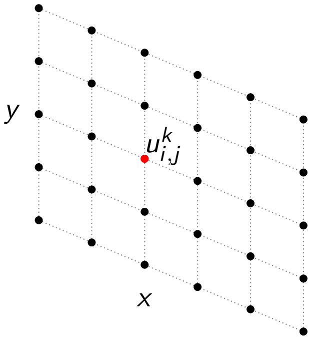
- $u^k_{i,j}$ represents the approximation of $u(x_i, y_j, t_k)$



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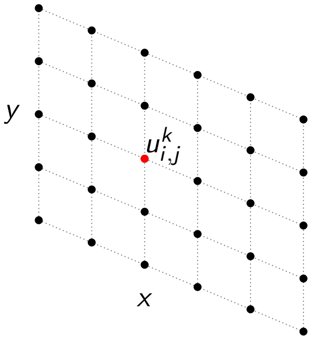
- $u_{i,j}^k$ represents the approximation of $u(x_i, y_j, t_k)$
- Space discretization: (x_i, y_j) where $x_i = i\Delta x$ and $y_j = j\Delta y$



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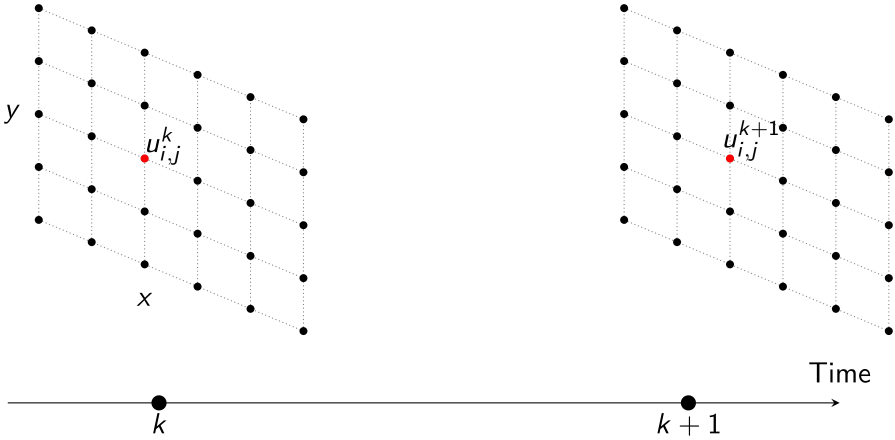

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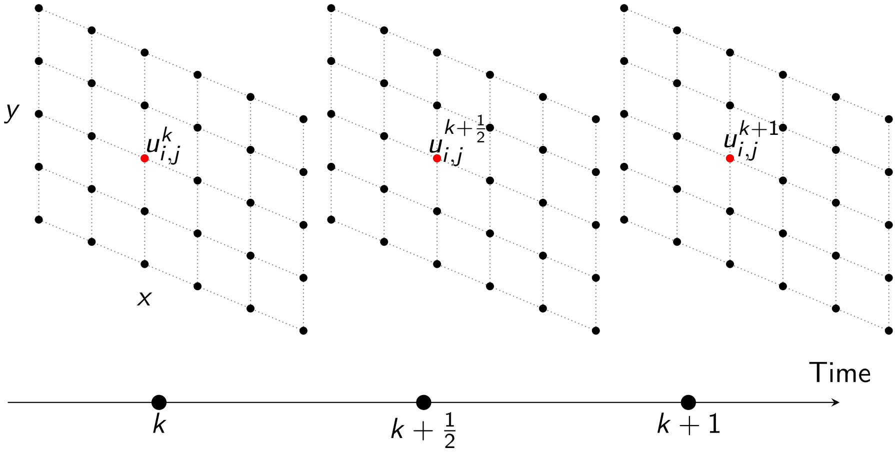

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$$\frac{u_{i,j}^{k+\frac{1}{2}} - u_{i,j}^{k}}{\Delta t/2} = D\left(\frac{u_{i+1,j}^{k+\frac{1}{2}} - 2u_{i,j}^{k+\frac{1}{2}} + u_{i-1,j}^{k+\frac{1}{2}}}{\Delta x^2} + \qquad\qquad\qquad\qquad\right)$$

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Simplify the equation for the first half-step

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Where only three terms are unknown: $u_{i-1,j}^{k+\frac{1}{2}}$, $u_{i,j}^{k+\frac{1}{2}}$, and $u_{i+1,j}^{k+\frac{1}{2}}$

First half-step (x implicit, y explicit)

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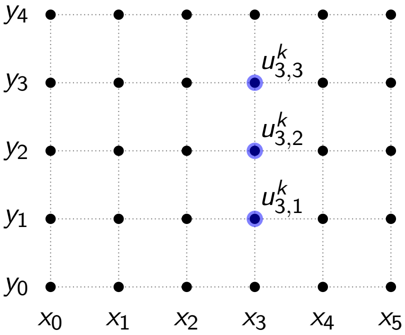

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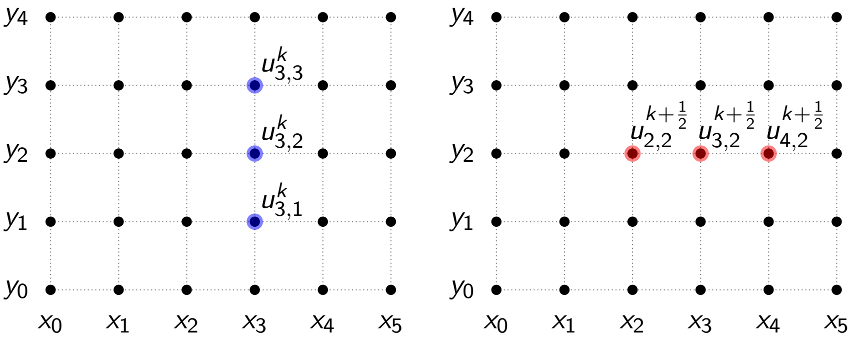
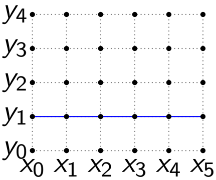


Writing the tridiagonal matrix

Let's fix $j = 1$ and write the equations for $i = 1, 2, 3, 4$:



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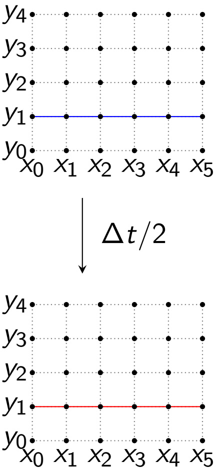

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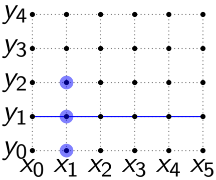
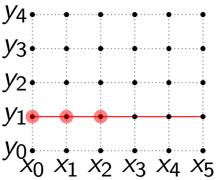


$\Delta t/2$



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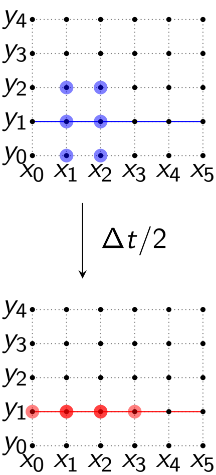

$$-\alpha u_{0,j}^{k+\frac{1}{2}} + (1+2\alpha)u_{1,j}^{k+\frac{1}{2}} - \alpha u_{2,j}^{k+\frac{1}{2}} = \alpha u_{1,j-1}^{k} + (1-2\alpha)u_{1,j}^{k} + \alpha u_{1,j+1}^{k}$$

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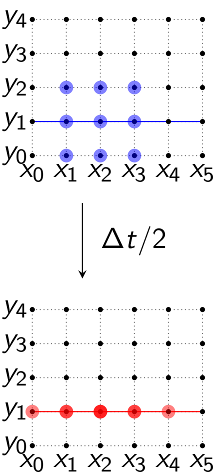

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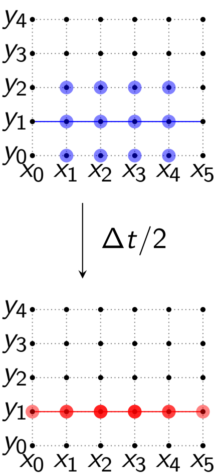

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$$-\alpha u_{3,j}^{k+\frac{1}{2}} + (1+2\alpha)u_{4,j}^{k+\frac{1}{2}} - \alpha u_{5,j}^{k+\frac{1}{2}} = \alpha u_{4,j-1}^{k} + (1-2\alpha)u_{4,j}^{k} + \alpha u_{4,j+1}^{k}$$

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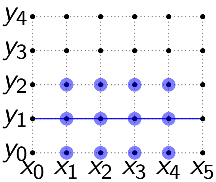
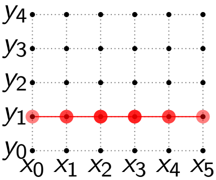


$$-\alpha u_{0,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{1,j}^{k+\frac{1}{2}} - \alpha u_{2,j}^{k+\frac{1}{2}} = \alpha u_{1,j-1}^{k} + (1-2\alpha) u_{1,j}^{k} + \alpha u_{1,j+1}^{k}$$
$$-\alpha u_{1,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{2,j}^{k+\frac{1}{2}} - \alpha u_{3,j}^{k+\frac{1}{2}} = \alpha u_{2,j-1}^{k} + (1-2\alpha) u_{2,j}^{k} + \alpha u_{2,j+1}^{k}$$
$$-\alpha u_{2,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{3,j}^{k+\frac{1}{2}} - \alpha u_{4,j}^{k+\frac{1}{2}} = \alpha u_{3,j-1}^{k} + (1-2\alpha) u_{3,j}^{k} + \alpha u_{3,j+1}^{k}$$
$$-\alpha u_{3,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{4,j}^{k+\frac{1}{2}} - \alpha u_{5,j}^{k+\frac{1}{2}} = \alpha u_{4,j-1}^{k} + (1-2\alpha) u_{4,j}^{k} + \alpha u_{4,j+1}^{k}$$

$\Delta t/2$

Which can be written as a matrix:



Writing the tridiagonal matrix

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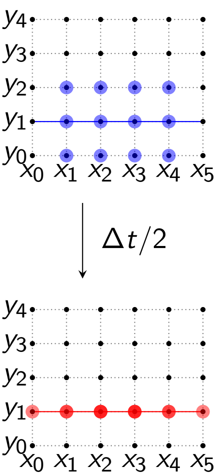


$$-\alpha u_{0,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{1,j}^{k+\frac{1}{2}} - \alpha u_{2,j}^{k+\frac{1}{2}} = \alpha u_{1,j-1}^{k} + (1-2\alpha) u_{1,j}^{k} + \alpha u_{1,j+1}^{k}$$

$$-\alpha u_{1,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{2,j}^{k+\frac{1}{2}} - \alpha u_{3,j}^{k+\frac{1}{2}} = \alpha u_{2,j-1}^{k} + (1-2\alpha) u_{2,j}^{k} + \alpha u_{2,j+1}^{k}$$

$$-\alpha u_{2,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{3,j}^{k+\frac{1}{2}} - \alpha u_{4,j}^{k+\frac{1}{2}} = \alpha u_{3,j-1}^{k} + (1-2\alpha) u_{3,j}^{k} + \alpha u_{3,j+1}^{k}$$

$$-\alpha u_{3,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{4,j}^{k+\frac{1}{2}} - \alpha u_{5,j}^{k+\frac{1}{2}} = \alpha u_{4,j-1}^{k} + (1-2\alpha) u_{4,j}^{k} + \alpha u_{4,j+1}^{k}$$

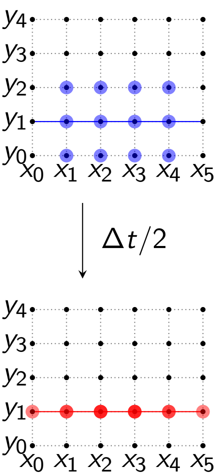
Which can be written as a matrix:

$$\begin{bmatrix} -\alpha & 1+2\alpha & -\alpha & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & -\alpha & 1+2\alpha & -\alpha \end{bmatrix} \begin{bmatrix} u_{0,j}^{k+\frac{1}{2}} \\ u_{1,j}^{k+\frac{1}{2}} \\ u_{2,j}^{k+\frac{1}{2}} \\ u_{3,j}^{k+\frac{1}{2}} \\ u_{4,j}^{k+\frac{1}{2}} \\ u_{5,j}^{k+\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} b_{1,j} \\ b_{2,j} \\ b_{3,j} \\ b_{4,j} \end{bmatrix}$$

Where $b_{i,j}$ is the right hand side of the equation, which is already known.

Writing the tridiagonal matrix

Let's fix $j = 1$ and write the equations for $i = 1, 2, 3, 4$:



$$-\alpha u_{0,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{1,j}^{k+\frac{1}{2}} - \alpha u_{2,j}^{k+\frac{1}{2}} = \alpha u_{1,j-1}^{k} + (1-2\alpha) u_{1,j}^{k} + \alpha u_{1,j+1}^{k}$$

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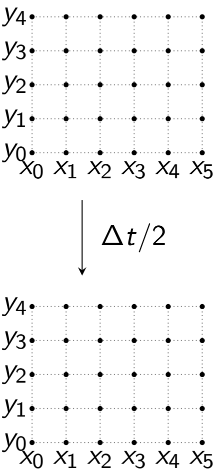
Where $b_{i,j}$ is the right hand side of the equation, which is already known.

Because of the boundary conditions, we can simplify the matrix to a tridiagonal matrix.

Writing the tridiagonal matrix

We get one system of equations for each j value.



Writing the tridiagonal matrix

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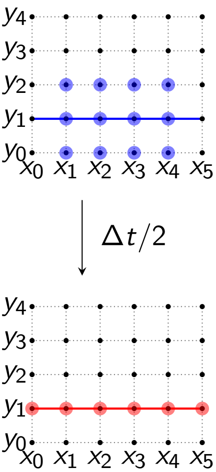
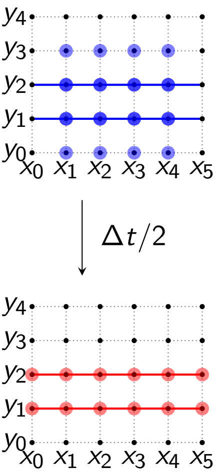


$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+\frac{1}{2}} \\ u_{2,1}^{k+\frac{1}{2}} \\ u_{3,1}^{k+\frac{1}{2}} \\ u_{4,1}^{k+\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{2,1} \\ b_{3,1} \\ b_{4,1} \end{bmatrix}$$

Writing the tridiagonal matrix

We get one system of equations for each j value.



$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+\frac{1}{2}} \\ u_{2,1}^{k+\frac{1}{2}} \\ u_{3,1}^{k+\frac{1}{2}} \\ u_{4,1}^{k+\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{2,1} \\ b_{3,1} \\ b_{4,1} \end{bmatrix}$$

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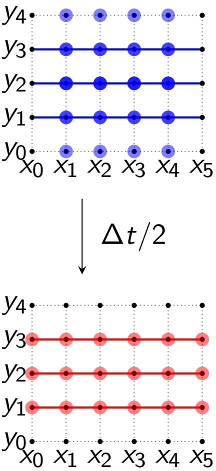
Writing the tridiagonal matrix

We get one system of equations for each j value.



$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+\frac{1}{2}} \\ u_{2,1}^{k+\frac{1}{2}} \\ u_{3,1}^{k+\frac{1}{2}} \\ u_{4,1}^{k+\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{2,1} \\ b_{3,1} \\ b_{4,1} \end{bmatrix}$$

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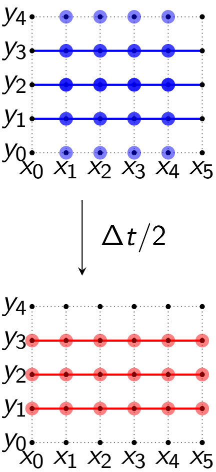
Writing the tridiagonal matrix

We get one system of equations for each j value.



$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+\frac{1}{2}} \\ u_{2,1}^{k+\frac{1}{2}} \\ u_{3,1}^{k+\frac{1}{2}} \\ u_{4,1}^{k+\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{2,1} \\ b_{3,1} \\ b_{4,1} \end{bmatrix}$$

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Now we can use the Thomas algorithm for solving each tridiagonal system.

That was the first half-step. Now we need to solve the second half-step.

Simplify the equation for the second half-step

$$\frac{u_{i,j}^{k+1} - u_{i,j}^{k+\frac{1}{2}}}{\Delta t/2} = D\left(\frac{u_{i+1,j}^{k+\frac{1}{2}} - 2u_{i,j}^{k+\frac{1}{2}} + u_{i-1,j}^{k+\frac{1}{2}}}{\Delta x^2} + \frac{u_{i,j+1}^{k+1} - 2u_{i,j}^{k+1} + u_{i,j-1}^{k+1}}{\Delta y^2}\right)$$

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Let's call $\alpha = \frac{D\Delta t}{2\Delta y^2}$

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Let's call $\alpha = \frac{D\Delta t}{2\Delta y^2}$

$$u_{i,j}^{k+1} - u_{i,j}^{k+\frac{1}{2}} = \alpha\left(u_{i+1,j}^{k+\frac{1}{2}} - 2u_{i,j}^{k+\frac{1}{2}} + u_{i-1,j}^{k+\frac{1}{2}} + u_{i,j+1}^{k+1} - 2u_{i,j}^{k+1} + u_{i,j-1}^{k+1}\right)$$

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After simplifying, we get:

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$$\boxed{-\alpha u_{i,j-1}^{k+1} + (1+2\alpha)u_{i,j}^{k+1} - \alpha u_{i,j+1}^{k+1} = \alpha u_{i-1,j}^{k+\frac{1}{2}} + (1-2\alpha)u_{i,j}^{k+\frac{1}{2}} + \alpha u_{i+1,j}^{k+\frac{1}{2}}}$$

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Where only three terms are unknown: $u_{i,j-1}^{k+1}$, $u_{i,j}^{k+1}$, and $u_{i,j+1}^{k+1}$

Second half-step (x explicit, y implicit)

$$\boxed{-\alpha u_{i,j-1}^{k+1} + (1+2\alpha)u_{i,j}^{k+1} - \alpha u_{i,j+1}^{k+1} = \alpha u_{i-1,j}^{k+\frac{1}{2}} + (1-2\alpha)u_{i,j}^{k+\frac{1}{2}} + \alpha u_{i+1,j}^{k+\frac{1}{2}}}$$

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Let's look around $i = 3$ and $j = 2$:

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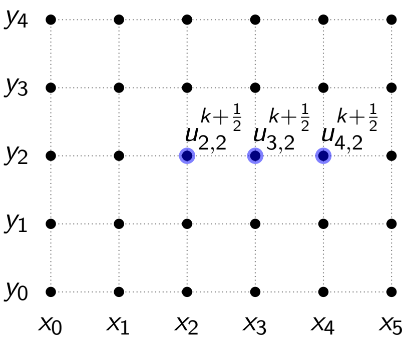

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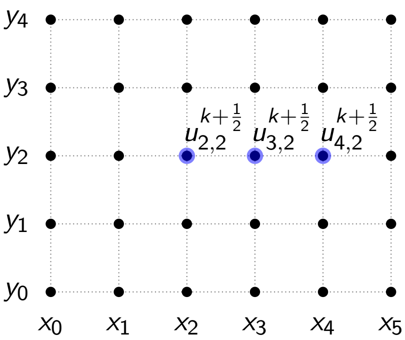

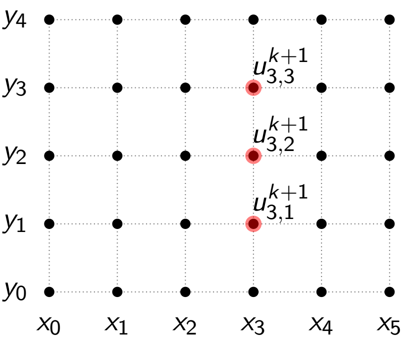

Writing the tridiagonal matrix

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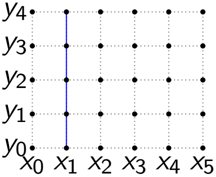

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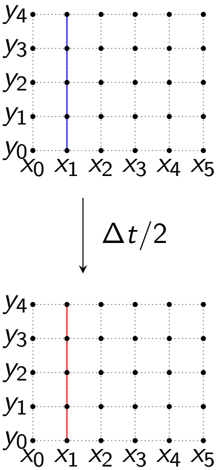

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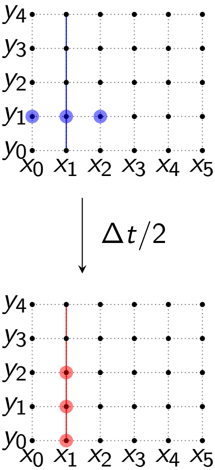

$$-\alpha u_{1,0}^{k+1} + (1+2\alpha) u_{1,1}^{k+1} - \alpha u_{1,2}^{k+1} = \alpha u_{0,1}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,1}^{k+\frac{1}{2}} + \alpha u_{2,1}^{k+\frac{1}{2}}$$

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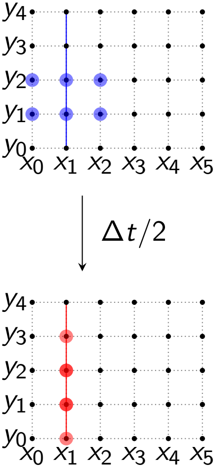

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$$-\alpha u_{1,1}^{k+1} + (1+2\alpha) u_{1,2}^{k+1} - \alpha u_{1,3}^{k+1} = \alpha u_{0,2}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,2}^{k+\frac{1}{2}} + \alpha u_{2,2}^{k+\frac{1}{2}}$$

Writing the tridiagonal matrix

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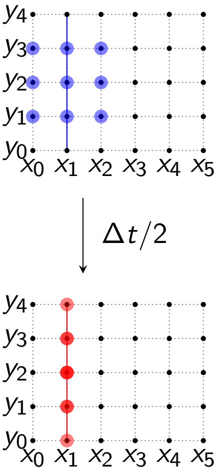

$$-\alpha u_{1,0}^{k+1} + (1+2\alpha) u_{1,1}^{k+1} - \alpha u_{1,2}^{k+1} = \alpha u_{0,1}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,1}^{k+\frac{1}{2}} + \alpha u_{2,1}^{k+\frac{1}{2}}$$

$$-\alpha u_{1,1}^{k+1} + (1+2\alpha) u_{1,2}^{k+1} - \alpha u_{1,3}^{k+1} = \alpha u_{0,2}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,2}^{k+\frac{1}{2}} + \alpha u_{2,2}^{k+\frac{1}{2}}$$

$$-\alpha u_{1,2}^{k+1} + (1+2\alpha) u_{1,3}^{k+1} - \alpha u_{1,4}^{k+1} = \alpha u_{0,3}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,3}^{k+\frac{1}{2}} + \alpha u_{2,3}^{k+\frac{1}{2}}$$

Writing the tridiagonal matrix

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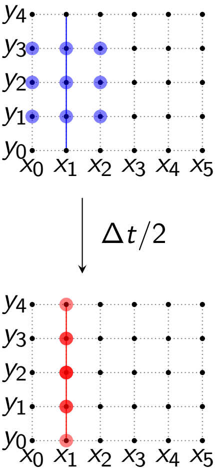


$$-\alpha u_{1,0}^{k+1} + (1+2\alpha) u_{1,1}^{k+1} - \alpha u_{1,2}^{k+1} = \alpha u_{0,1}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,1}^{k+\frac{1}{2}} + \alpha u_{2,1}^{k+\frac{1}{2}}$$

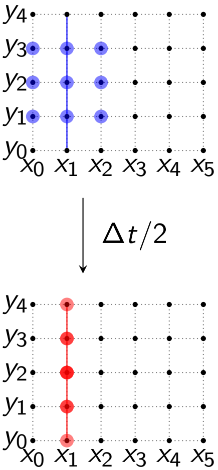
$$-\alpha u_{1,1}^{k+1} + (1+2\alpha) u_{1,2}^{k+1} - \alpha u_{1,3}^{k+1} = \alpha u_{0,2}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,2}^{k+\frac{1}{2}} + \alpha u_{2,2}^{k+\frac{1}{2}}$$

$$-\alpha u_{1,2}^{k+1} + (1+2\alpha) u_{1,3}^{k+1} - \alpha u_{1,4}^{k+1} = \alpha u_{0,3}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,3}^{k+\frac{1}{2}} + \alpha u_{2,3}^{k+\frac{1}{2}}$$

Which can be written as a matrix:

Writing the tridiagonal matrix

Let's fix $i = 1$ and write the equations for $j = 1, 2, 3$:



$$-\alpha u_{1,0}^{k+1} + (1+2\alpha)u_{1,1}^{k+1} - \alpha u_{1,2}^{k+1} = \alpha u_{0,1}^{k+\frac{1}{2}} + (1-2\alpha)u_{1,1}^{k+\frac{1}{2}} + \alpha u_{2,1}^{k+\frac{1}{2}}$$
$$-\alpha u_{1,1}^{k+1} + (1+2\alpha)u_{1,2}^{k+1} - \alpha u_{1,3}^{k+1} = \alpha u_{0,2}^{k+\frac{1}{2}} + (1-2\alpha)u_{1,2}^{k+\frac{1}{2}} + \alpha u_{2,2}^{k+\frac{1}{2}}$$
$$-\alpha u_{1,2}^{k+1} + (1+2\alpha)u_{1,3}^{k+1} - \alpha u_{1,4}^{k+1} = \alpha u_{0,3}^{k+\frac{1}{2}} + (1-2\alpha)u_{1,3}^{k+\frac{1}{2}} + \alpha u_{2,3}^{k+\frac{1}{2}}$$

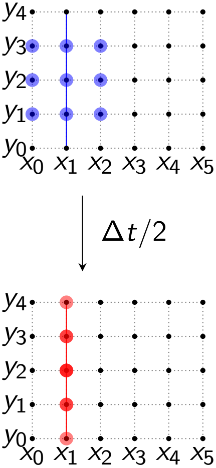
Which can be written as a matrix:

$$\begin{bmatrix} -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha \end{bmatrix} \begin{bmatrix} u_{1,0}^{k+1} \\ u_{1,1}^{k+1} \\ u_{1,2}^{k+1} \\ u_{1,3}^{k+1} \\ u_{1,4}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{1,2} \\ b_{1,3} \end{bmatrix}$$

Where $b_{i,j}$ is the right hand side of the equation, which is already known.

Writing the tridiagonal matrix

Let's fix $i = 1$ and write the equations for $j = 1, 2, 3$:



$$-\alpha u_{1,0}^{k+1} + (1+2\alpha) u_{1,1}^{k+1} - \alpha u_{1,2}^{k+1} = \alpha u_{0,1}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,1}^{k+\frac{1}{2}} + \alpha u_{2,1}^{k+\frac{1}{2}}$$

$$-\alpha u_{1,1}^{k+1} + (1+2\alpha) u_{1,2}^{k+1} - \alpha u_{1,3}^{k+1} = \alpha u_{0,2}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,2}^{k+\frac{1}{2}} + \alpha u_{2,2}^{k+\frac{1}{2}}$$

$$-\alpha u_{1,2}^{k+1} + (1+2\alpha) u_{1,3}^{k+1} - \alpha u_{1,4}^{k+1} = \alpha u_{0,3}^{k+\frac{1}{2}} + (1-2\alpha) u_{1,3}^{k+\frac{1}{2}} + \alpha u_{2,3}^{k+\frac{1}{2}}$$

Which can be written as a matrix:

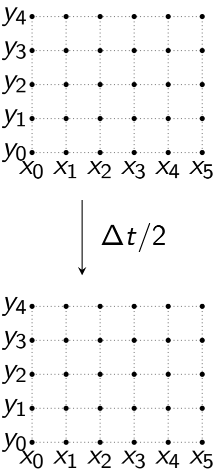
$$\begin{bmatrix} -\alpha & 1+2\alpha & -\alpha & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & -\alpha & 1+2\alpha & -\alpha \end{bmatrix} \begin{bmatrix} u_{1,0}^{k+1} \\ u_{1,1}^{k+1} \\ u_{1,2}^{k+1} \\ u_{1,3}^{k+1} \\ u_{1,4}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{1,2} \\ b_{1,3} \end{bmatrix}$$

Where $b_{i,j}$ is the right hand side of the equation, which is already known.

Because of the boundary conditions, we can simplify the matrix to a tridiagonal matrix.

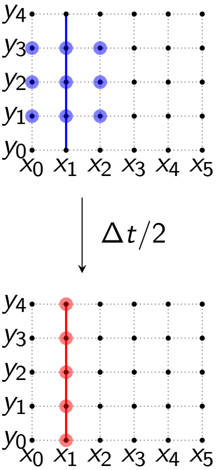
Writing the tridiagonal matrix

We get one system of equations for each i value.



Writing the tridiagonal matrix

We get one system of equations for each i value.



$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+1} \\ u_{1,2}^{k+1} \\ u_{1,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{1,2} \\ b_{1,3} \end{bmatrix}$$

Writing the tridiagonal matrix

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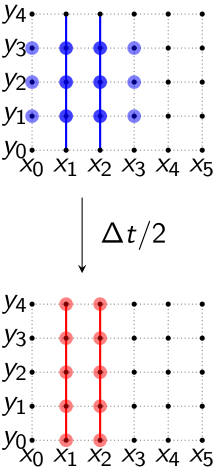
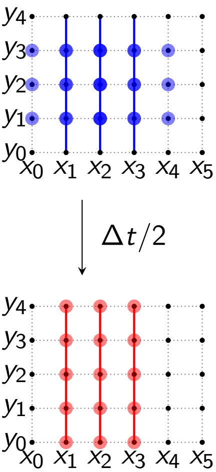


$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+1} \\ u_{1,2}^{k+1} \\ u_{1,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{1,2} \\ b_{1,3} \end{bmatrix}$$

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{2,1}^{k+1} \\ u_{2,2}^{k+1} \\ u_{2,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{bmatrix}$$

Writing the tridiagonal matrix

We get one system of equations for each i value.

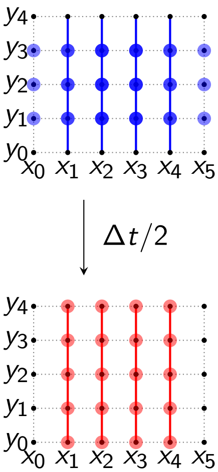


$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+1} \\ u_{1,2}^{k+1} \\ u_{1,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{1,2} \\ b_{1,3} \end{bmatrix}$$

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{2,1}^{k+1} \\ u_{2,2}^{k+1} \\ u_{2,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{bmatrix}$$

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{3,1}^{k+1} \\ u_{3,2}^{k+1} \\ u_{3,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{3,1} \\ b_{3,2} \\ b_{3,3} \end{bmatrix}$$

We get one system of equations for each i value.



$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{1,1}^{k+1} \\ u_{1,2}^{k+1} \\ u_{1,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{1,1} \\ b_{1,2} \\ b_{1,3} \end{bmatrix}$$

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{2,1}^{k+1} \\ u_{2,2}^{k+1} \\ u_{2,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{bmatrix}$$

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{3,1}^{k+1} \\ u_{3,2}^{k+1} \\ u_{3,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{3,1} \\ b_{3,2} \\ b_{3,3} \end{bmatrix}$$

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 \\ -\alpha & 1+2\alpha & -\alpha \\ 0 & -\alpha & 1+2\alpha \end{bmatrix} \begin{bmatrix} u_{4,1}^{k+1} \\ u_{4,2}^{k+1} \\ u_{4,3}^{k+1} \end{bmatrix} = \begin{bmatrix} b_{4,1} \\ b_{4,2} \\ b_{4,3} \end{bmatrix}$$

Now we can use the Thomas algorithm for solving each individual tridiagonal system.

Adding point-like sinks and sources

$$\frac{\partial u}{\partial t} = D\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + K_{out}\sum_{p=1}^{N}\delta(r_p - r) - K_{in}\sum_{c=1}^{N}\delta(r_c - r)$$

Adding point-like sinks and sources

$$\frac{\partial u}{\partial t} = D\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + K_{out}\sum_{p=1}^{N}\delta(r_p - r) - K_{in}\sum_{c=1}^{N}\delta(r_c - r)$$

We can discretize as follows:

$$\frac{u_{i,j}^{k+\frac{1}{2}} - u_{i,j}^{k}}{\Delta t/2} = D\left(\frac{u_{i+1,j}^{k+\frac{1}{2}} - 2u_{i,j}^{k+\frac{1}{2}} + u_{i-1,j}^{k+\frac{1}{2}}}{\Delta x^2} + \frac{u_{i,j+1}^{k} - 2u_{i,j}^{k} + u_{i,j-1}^{k}}{\Delta y^2}\right)$$

Adding point-like sinks and sources

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$$+ K_{out}\sum_{p=1}^{N}\delta_\varepsilon(r_p - r_{i,j}) - K_{in}\sum_{c=1}^{N}\delta_\varepsilon(r_c - r_{i,j}).$$

Adding point-like sinks and sources

$$\frac{\partial u}{\partial t} = D\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + K_{out}\sum_{p=1}^{N}\delta(r_p - r) - K_{in}\sum_{c=1}^{N}\delta(r_c - r)$$

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$$+ K_{out}\sum_{p=1}^{N}\delta_\varepsilon(r_p - r_{i,j}) - K_{in}\sum_{c=1}^{N}\delta_\varepsilon(r_c - r_{i,j}).$$

Adding point-like sinks and sources

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$$+ K_{out}\sum_{p=1}^{N}\delta_\varepsilon(r_p - r_{i,j}) - K_{in}\sum_{c=1}^{N}\delta_\varepsilon(r_c - r_{i,j}).$$

We define $\alpha = \frac{D\Delta t}{2\Delta x^2}$

Adding point-like sinks and sources

$$\frac{\partial u}{\partial t} = D\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + K_{out}\sum_{p=1}^{N}\delta(r_p - r) - K_{in}\sum_{c=1}^{N}\delta(r_c - r)$$

We can discretize as follows:

$$\frac{u_{i,j}^{k+\frac{1}{2}} - u_{i,j}^{k}}{\Delta t/2} = D\left(\frac{u_{i+1,j}^{k+\frac{1}{2}} - 2u_{i,j}^{k+\frac{1}{2}} + u_{i-1,j}^{k+\frac{1}{2}}}{\Delta x^2} + \frac{u_{i,j+1}^{k} - 2u_{i,j}^{k} + u_{i,j-1}^{k}}{\Delta y^2}\right) + K_{out}\sum_{p=1}^{N}\delta_\varepsilon(r_p - r_{i,j}) - K_{in}\sum_{c=1}^{N}\delta_\varepsilon(r_c - r_{i,j}).$$

We define $\alpha = \frac{D\Delta t}{2\Delta x^2}$

$$\boxed{-\alpha u_{i-1,j}^{k+\frac{1}{2}} + (1+2\alpha)u_{i,j}^{k+\frac{1}{2}} - \alpha u_{i+1,j}^{k+\frac{1}{2}} = \alpha u_{i,j-1}^{k} + (1-2\alpha)u_{i,j}^{k} + \alpha u_{i,j+1}^{k} + K_{out}\sum_{p=1}^{N}\delta_\varepsilon(r_p - r_{i,j}) - K_{in}\sum_{c=1}^{N}\delta_\varepsilon(r_p - r_{i,j})}$$

$$\boxed{-\alpha u_{i-1,j}^{k+\frac{1}{2}} + (1+2\alpha) u_{i,j}^{k+\frac{1}{2}} - \alpha u_{i+1,j}^{k+\frac{1}{2}} = \alpha u_{i,j-1}^{k} + (1-2\alpha) u_{i,j}^{k} + \alpha u_{i,j+1}^{k} + K_{out} \sum_{p=1}^{N} \delta_\varepsilon(r_p - r_{i,j}) - K_{in} \sum_{c=1}^{N} \delta_\varepsilon(r_p - r_{i,j})}$$

On the steady state, $\frac{\partial u}{\partial t} = 0$, which means that the difference between the two half-steps is zero: $\Delta u = u_{i,j}^{k+1/2} - u_{i,j}^{k} = 0$. We can then write the equation as:

$$\boxed{-\alpha u_{i-1,j}^{k+\frac{1}{2}} - \alpha u_{i+1,j}^{k+\frac{1}{2}} = \alpha u_{i,j-1}^{k} - 4\alpha u_{i,j}^{k} + \alpha u_{i,j+1}^{k} + K_{out} \sum_{p=1}^{N} \delta_\varepsilon(r_p - r_{i,j}) - K_{in} \sum_{c=1}^{N} \delta_\varepsilon(r_p - r_{i,j})}$$